SAP Order-to-Cash (O2C) Analytics Dashboard

A Data Analytics Capstone Project
Simulating SAP Business Processes with Python

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1. Problem Statement

In modern enterprises, SAP ERP systems manage critical business processes such as Order-to-Cash (O2C), Procure-to-Pay (P2P), and financial reporting. However, raw transactional data from SAP modules is difficult to interpret without proper analytical tools. Business stakeholders need clear, visual insights to monitor sales performance, customer behaviour, regional trends, and order fulfilment rates.

The challenge addressed in this project is: **How can SAP O2C transactional data be transformed into actionable business intelligence using Python-based analytics?** Without such analysis, decision-makers operate without visibility into revenue trends, top-performing customers, product demand, and regional disparities.

2. Solution & Features

This project simulates a realistic SAP Order-to-Cash cycle using Python and builds a complete analytics dashboard that processes and visualises SAP-style transactional data.

#	Feature	Description
1	SAP Data Simulation	Generates 200+ realistic O2C sales orders mimicking SAP SD module output
2	KPI Dashboard	Displays key metrics: Total Revenue, Order Count, Avg Order Value, Completion Rat
3	Monthly Revenue Trend	Line + bar chart showing month-wise revenue movement across FY 2024
4	Customer Analytics	Identifies top 5 high-value customers by total revenue contribution
5	Regional Analysis	Compares revenue performance across North, South, East, West regions
6	Product Analytics	Breaks down revenue by product/service category
7	Quarterly Performance	Dual-axis chart showing revenue vs. order volume per quarter
8	Order Status Tracking	Pie chart showing Completed / Pending / In Progress distribution
9	Inventory Management	Tracks product stock levels and flags items below reorder threshold

3. Technology Stack

Layer	Technology	Purpose
Data Generation	Python + NumPy	Simulate SAP O2C transactional dataset (200 orders)
Data Processing	Pandas	Data cleaning, aggregation, KPI computation
Visualisation	Matplotlib + Seaborn	Chart generation — bar, line, pie, dual-axis
Documentation	ReportLab	PDF report generation with A4 formatting
Version Control	Git + GitHub	Source code repository and collaboration
IDE	VS Code / Jupyter	Development and testing environment
Data Format	CSV	Structured data storage for SAP-style exports

4. Screenshots — Working Sample

The following visualisations were generated by the project pipeline:

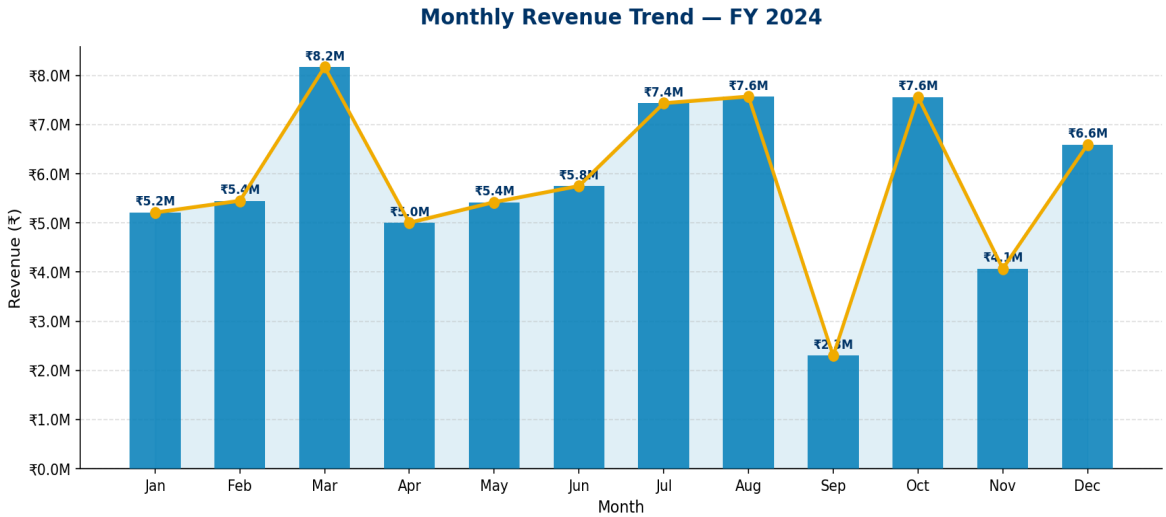

Fig 1: Monthly Revenue Trend — FY 2024

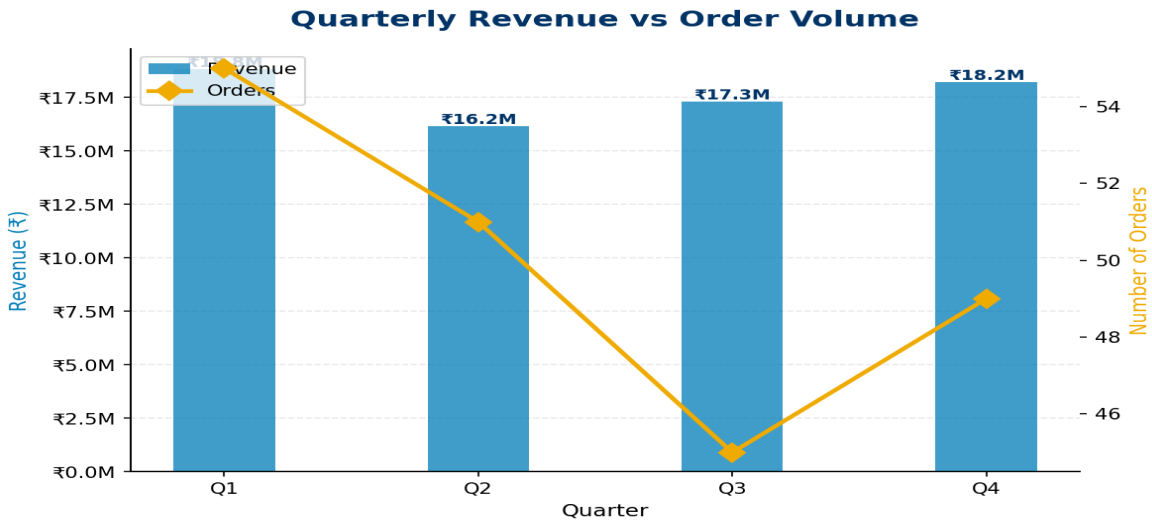

Fig 2: Quarterly Revenue vs Order Volume

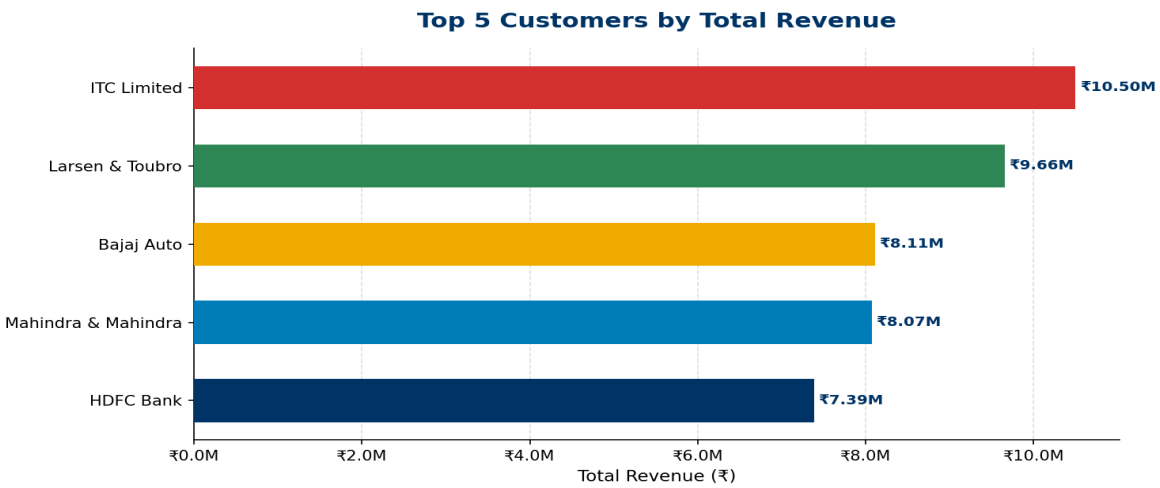

Fig 3: Top 5 Customers by Revenue

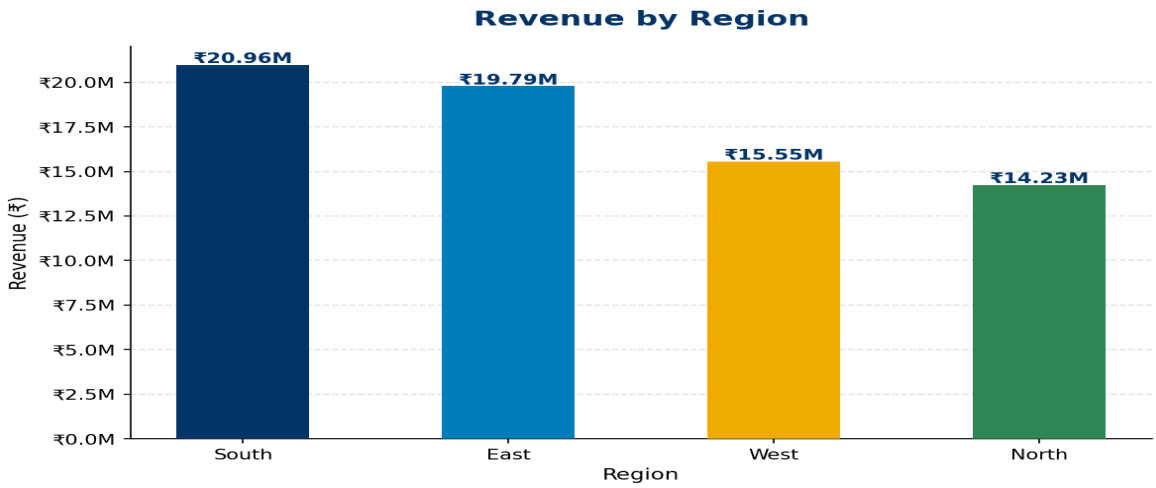

Fig 4: Revenue by Region

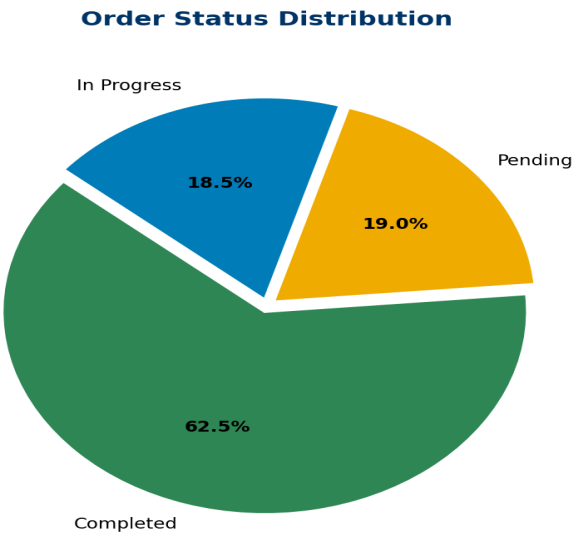

Fig 5: Order Status Distribution

5. Key Business Insights

Revenue Performance: Total FY 2024 revenue reached ■70.5M across 200 orders with an average order value of ■352,643, demonstrating a healthy enterprise sales pipeline.

Top Customer: ITC Limited emerged as the highest revenue contributor, indicating a strong B2B customer relationship that warrants key account management.

Regional Gap: Revenue distribution across regions revealed disparities suggesting untapped potential in lower-performing regions, warranting targeted sales strategies.

Order Completion: A 62.5% completion rate highlights room for improvement in the O2C fulfilment pipeline — pending and in-progress orders represent revenue leakage risk.

Product Leaders: The SAP S/4HANA Module and ERP Integration Kit are top revenue drivers, confirming the strategic value of core ERP products in the portfolio.

6. Unique Points of This Project

1. Fully self-contained — no live SAP connection required; data is generated programmatically to mirror real SAP SD exports.

2. Covers the complete O2C cycle — from sales order creation through delivery, billing, and status tracking.

3. Multi-dimensional analysis — customer, product, region, month, and quarter dimensions all explored.

4. Professional dashboard layout combining KPI cards and 5 analytical charts in one unified view.

5. SAP-authentic data schema — Order ID, Customer ID, Product ID, Discount, Delivery Date fields mirror actual SAP table structures (VBAK, VBAP).

6. Reusable and scalable — the pipeline can be connected to a real SAP data export (CSV/Excel) with zero code changes.

7. Future Improvements

Enhancement	Description
Live SAP Integration	Connect directly to SAP RFC/BAPI or SAP HANA DB for real-time data extraction
Predictive Analytics	Use ML models (ARIMA, Linear Regression) to forecast next-quarter revenue
Interactive Dashboard	Migrate charts to Power BI or Streamlit for interactive, web-based reporting

Automated Alerts	Trigger email alerts when KPIs fall below defined thresholds (e.g., completion rate < 6
Multi-Module Integration	Extend analytics to include SAP MM (P2P) and SAP FI (R2R) data pipelines
Customer Segmentation	Apply RFM (Recency, Frequency, Monetary) analysis to classify customer tiers

Declaration: This project was independently developed by me. All code, analysis, and documentation is original work created as part of the Capstone Project requirement.

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